A plane on a sextic Calabi–Yau fourfold and its residual quintics

Benjamin Stanley Frohman
Independent researcher

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Abstract

Let

$$F = x_0^5x_3 + x_3^6 + x_1^5x_4 + x_4^6 + x_2^5x_5 + x_5^6, \qquad X = V(F) \subset \mathbb{P}^5, \qquad \Pi = V(x_3, x_4, x_5).$$

Then X is a smooth Calabi–Yau fourfold containing the coordinate plane Π. Projection from Π yields a family of residual quintic surfaces $S \subset X$. For a general parameter the surface S is smooth of general type, $K_S \simeq \mathcal{O}_S(1)$, and $[S] = h^2 - [\Pi]$ in $H^4(X)$. Intersection numbers on X are computed from the normal bundle of Π. The note records what this geometry does and does not imply: it is not a Fourier–Mukai partner fourfold, not a disproof of the Hodge conjecture, and not a Clay counterexample.

1 The host

Write $h = c_1(\mathcal{O}_X(1))$. The polynomial F is homogeneous of degree 6, so X is a sextic hypersurface in $\mathbb{P}^5$. Adjunction gives

$$K_X = \mathcal{O}_X(6 - 5 - 1) = \mathcal{O}_X.$$

The partials of F vanish simultaneously only at the origin in $\mathbb{A}^6$, so X is smooth. Thus X is a Calabi–Yau fourfold. Standard Hodge numbers of a smooth sextic fourfold are

$$h^{4,0} = 1, \qquad h^{3,1} = 426, \qquad h^{2,2} = 1752.$$

The Lefschetz hyperplane theorem gives $\mathrm{Pic}(X) = \mathbb{Z}h$. The plane Π lies on X because $F \in \langle x_3, x_4, x_5\rangle$. Ledger:

$$I(X) = \langle F\rangle, \qquad I(\Pi) = \langle x_3, x_4, x_5\rangle.$$

The class $[\Pi]$ is algebraic. This is the easy Hodge arrow for one class on one fourfold, not the statement $\forall X\, \forall \gamma\, \exists Z_i$.

2 Residual quintics

A point $[a : b : c] \in \mathbb{P}^2$ determines the linear $\mathbb{P}^3$

$$L_{a,b,c} = \big\{[x_0 : x_1 : x_2 : ta : tb : tc]\big\}.$$

Its ideal in $\mathbb{P}^5$ is the 2×2 minors of $\begin{pmatrix} x_3 & x_4 & x_5 \\ a & b & c \end{pmatrix}$:

$$I(L) = \langle bx_3 - ax_4,\ cx_3 - ax_5,\ cx_4 - bx_5 \rangle.$$

Restriction of F factors as

$$F\big|_L = t\big(ax_0^5 + bx_1^5 + cx_2^5 + t^5(a^6 + b^6 + c^6)\big).$$

Write $\mu = a^6 + b^6 + c^6$ and

$$R_{a,b,c} = ax_0^5 + bx_1^5 + cx_2^5 + t^5\mu \in \mathbb{C}[x_0, x_1, x_2, t]_5, \qquad S = V(R) \subset L.$$

The factor $t = 0$ is Π. The residual is S. Thus $X \cap L = \Pi \cup S$ and $S \subset X$.

Theorem 1 (Residual quintics of (F, Π))**.** *If $abc\mu \neq 0$, then S is a smooth quintic surface in $L \cong \mathbb{P}^3$, of general type, with $K_S \simeq \mathcal{O}_S(1)$ very ample, and*

$$[S] = h^2 - [\Pi] \quad \textit{in } H^4(X).$$

Proof. The identity $F|_L = tR$ is polynomial. The partials of R are

$$5ax_0^4, \quad 5bx_1^4, \quad 5cx_2^4, \quad 5t^4\mu.$$

If $abc\mu \neq 0$ these vanish only at the origin, so S is smooth. Adjunction in $\mathbb{P}^3$ gives $K_S = \mathcal{O}_S(5-4) = \mathcal{O}_S(1)$. That line bundle is the hyperplane class of the given embedding, hence very ample, and S is of general type. A linear $\mathbb{P}^3$ cuts X in class h^2. Additivity of cycles on $X \cap L = \Pi \cup S$ gives $[S] = h^2 - [\Pi]$. □

If $\mu = 0$ and $abc \neq 0$, then R is a cone, singular at $[0:0:0:1]$. The discriminant in the base therefore contains the Fermat sextic $a^6 + b^6 + c^6 = 0$ (genus 10).

3 Intersections

Smooth inclusions $\Pi \subset X \subset \mathbb{P}^5$ give

$$0 \to N_{\Pi/X} \to N_{\Pi/\mathbb{P}^5} \to N_{X/\mathbb{P}^5}\big|_\Pi \to 0,$$

with $N_{\Pi/\mathbb{P}^5} = \mathcal{O}_\Pi(1)^{\oplus 3}$ and $N_{X/\mathbb{P}^5}|_\Pi = \mathcal{O}_\Pi(6)$. On Π,

$$\frac{\partial F}{\partial x_3} = x_0^5, \quad \frac{\partial F}{\partial x_4} = x_1^5, \quad \frac{\partial F}{\partial x_5} = x_2^5,$$

so the map $\mathcal{O}(1)^3 \to \mathcal{O}(6)$ is $(s_3, s_4, s_5) \mapsto x_0^5 s_3 + x_1^5 s_4 + x_2^5 s_5$, surjective on Π. Whitney:

$$c(N_{\Pi/X}) = \frac{(1+L)^3}{1+6L} = (1 + 3L + 3L^2)(1 - 6L + 36L^2) = 1 - 3L + 21L^2$$

on $\mathbb{P}^2$. Hence $c_1(N) = -3L$ and $c_2(N) = 21$. For a surface in a fourfold, $Z^2 = \int_Z c_2(N_{Z/X})$, so $\Pi^2 = 21$. Combined with $h^4 = 6$ and $h^2 \cdot [\Pi] = 1$,

$$\begin{aligned} [S] \cdot h^2 &= 6 - 1 = 5, \\ [S] \cdot [\Pi] &= 1 - 21 = -20, \\ [S]^2 &= 6 - 2 + 21 = 25. \end{aligned}$$

The curve $C = \Pi \cap S$ is $ax_0^5 + bx_1^5 + cx_2^5 = 0 \subset \Pi \cong \mathbb{P}^2$. If $abc \neq 0$ it is a smooth plane quintic of genus 6.

A smooth quintic surface in $\mathbb{P}^3$ has $K_S^2 = 5$. From

$$0 \to \mathcal{O}_{\mathbb{P}^3}(-5) \to \mathcal{O}_{\mathbb{P}^3} \to \mathcal{O}_S \to 0$$

one has $\chi(\mathcal{O}_{\mathbb{P}^3}) = 1$ and, by Serre duality, $\chi(\mathcal{O}(-5)) = -4$, so $\chi(\mathcal{O}_S) = 5$. Then $p_g = h^0(\mathcal{O}_S(1)) = 4$, $q = 0$, Noether gives $e(S) = 55$, and $b_2 = 53$, $h^{1,1} = 45$.

These classes remain in the span of $\{h^2, [\Pi]\}$. No new independent Hodge class is produced.

4 Derived categories

Write $D^b(Z)$ for the bounded derived category of coherent sheaves. Orlov's theorem for a Calabi–Yau hypersurface supplies

$$D^b(\operatorname{coh} X) \;\simeq\; \mathrm{HMF}^{\mathrm{gr}}(F).$$

This is an equivalence of 4-Calabi–Yau categories attached to the same polynomial. It is not a second fourfold.

A derived equivalence $D^b(A) \simeq D^b(B)$ of smooth projective varieties is, by Orlov representability, Fourier–Mukai: there is a kernel $\mathcal{E} \in D^b(A \times B)$. Bondal–Orlov reconstruction says that if K or K^{-1} is ample then the variety is recovered from D^b. Thus $D^b(S)$ reconstructs S, while $D^b(X)$ does not reconstruct X because $K_X \simeq \mathcal{O}$. Moreover $\dim S = 2 \neq 4 = \dim X$, and the Serre functors are $(- \otimes K_S)[2]$ and $[4]$ respectively. So S is not a Fourier–Mukai partner of X.

The blowup $\mathrm{Bl}_\Pi X$ has $K = 2E \not\simeq \mathcal{O}$. It is not Calabi–Yau and is not a partner.

Conjecture 1 (FM partner of $V(F)$)**.** *There exist a smooth projective fourfold $Y \not\simeq X$ with $\omega_Y \simeq \mathcal{O}_Y$ and a kernel $\mathcal{E} \in D^b(X \times Y)$ such that $\Phi_{\mathcal{E}}$ is an equivalence $D^b(X) \xrightarrow{\sim} D^b(Y)$.*

This remains open. A partner is a variety plus a kernel plus a proof. None of those three is written for this F.

5 What this note is not

Hodge on fourfolds asks: for every smooth projective fourfold and every class $\gamma \in H^{2,2} \cap H^4(\mathbb{Q})$, do surfaces Z_i exist with $\gamma = \sum a_i [Z_i]$? A Clay counterexample is one named fourfold and one named Hodge class with a proof that no such surfaces exist.

On this X, $[\Pi]$ and $[S]$ are algebraic. Those pairs $(X, [\Pi])$ and $(X, [S])$ are instances of the conjecture, not witnesses against it. A missing Lean instance is neither a proof of Hodge nor a counterexample. The blowup $\mathrm{Bl}_\Pi X$ is a different fourfold; classes there do not disprove Hodge on X.

Derived equivalences do not turn $[\Pi]$ or $[S]$ into Mukai vectors. This host has $h^{3,1} = 426$ and no Kuznetsov K3 component.

6 Ledger

$X = V(F)$	locked CY fourfold
Π	locked plane on X
$S = S_{a,b,c}$	residual quintic surface on X
Y	unwritten
$D^b(X) \simeq \mathrm{HMF}^{\mathrm{gr}}(F)$	same host

Companion files: `BenFrohman/DerivedCategories`, draft of 23 September 2026.